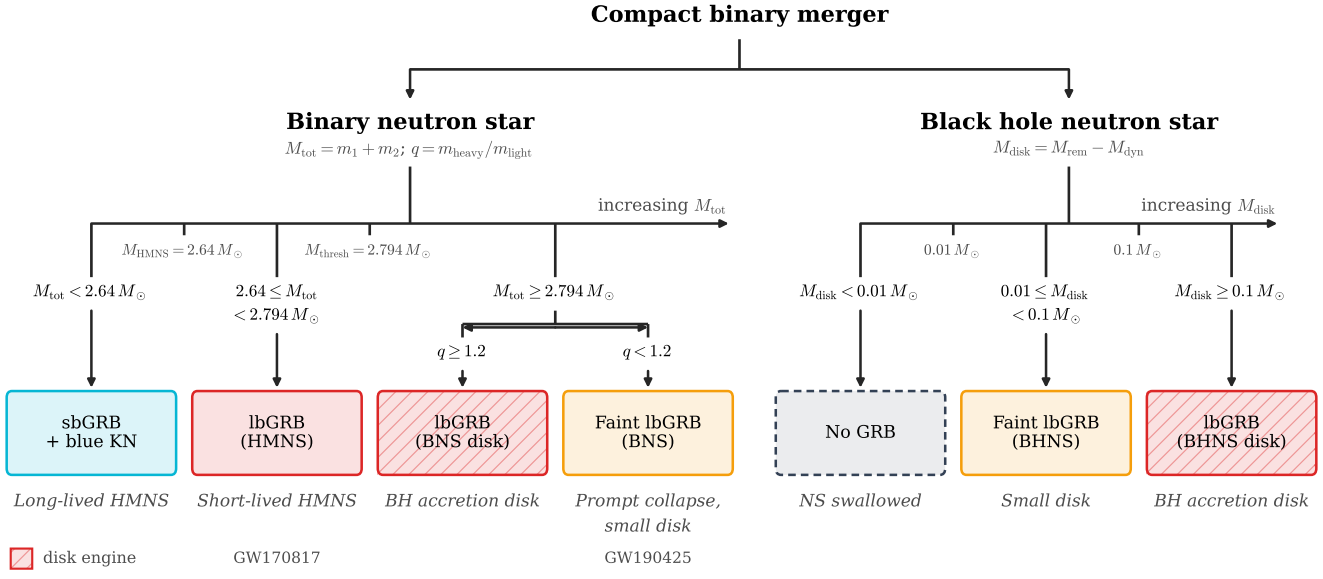

Compact binary merger
Binary neutron star
$M_{\rm tot} = m_1 + m_2$; $q = m_{\rm heavy}/m_{\rm light}$
increasing $M_{\rm tot}$
$M_{\rm HMNS} = 2.64\,M_\odot$
$M_{\rm thresh} = 2.794\,M_\odot$
$M_{\rm tot} < 2.64\,M_\odot$
$2.64 \le M_{\rm tot} < 2.794\,M_\odot$
$M_{\rm tot} \ge 2.794\,M_\odot$
$q \ge 1.2$
$q < 1.2$
sbGRB + blue KN
lbGRB (HMNS)
lbGRB (BNS disk)
Faint lbGRB (BNS)
Long-lived HMNS
Short-lived HMNS
BH accretion disk
Prompt collapse, small disk
Black hole neutron star
$M_{\rm disk} = M_{\rm rem} - M_{\rm dyn}$
increasing $M_{\rm disk}$
$0.01\,M_\odot$
$0.1\,M_\odot$
$M_{\rm disk} < 0.01\,M_\odot$
$0.01 \le M_{\rm disk} < 0.1\,M_\odot$
$M_{\rm disk} \ge 0.1\,M_\odot$
No GRB
Faint lbGRB (BHNS)
lbGRB (BHNS disk)
NS swallowed
Small disk
BH accretion disk
disk engine
GW170817
GW190425